

REQUEST FOR AUTHORIZATION

Pilot Deployment of a Shared-Laboratory Endpoint Security Configuration and Associated Measurement Study

To:

The Director, Information Technology Office
North South University, Bashundhara, Dhaka

Through:

Dr. Nova Ahmed, Department of Computer Science and Engineering, Faculty Supervisor, DIAL Lab

From:

Khondokar Sajid, ID 2211954042, Department of Computer Science and Engineering, North South University

Date: _____

Subject: Request for authorization to conduct a supervised, zero-cost security pilot and measurement study on one computer laboratory.

Respected Sir/Madam,

I am a final-semester student in the Department of Computer Science and Engineering, working under the supervision of Dr. Nova Ahmed in the DIAL Lab. I am writing to request authorization to carry out a supervised research pilot aimed at identifying and closing a security gap that currently affects the University's shared computer laboratories. This letter sets out the purpose of the work, exactly what I am requesting permission to do, the safeguards that will be observed, and the limited and reversible nature of every action involved.

1. Purpose and Justification

Shared laboratory computers at the University run the Windows operating system and the Chrome or Edge browser in their default configuration. In this configuration, a student's saved passwords, synchronised personal data, and active sessions can remain accessible to the next person who uses the same machine, without any malware, software exploit, or password cracking. This represents a genuine risk of credential exposure and privacy compromise for students and staff, and may constitute a reportable data-protection concern under the Personal Data Protection Ordinance 2025. The purpose of this pilot is to measure the extent of this exposure on University machines and to demonstrate that it can be closed using only configuration changes and free software already available to the University, at no additional licensing cost.

2. Specific Authorization Requested

I respectfully request authorization, under the supervision of the IT Office and Dr. Nova Ahmed, to perform the following on one designated laboratory (hereafter "the pilot laboratory") and one comparable laboratory used only for before-and-after comparison:

- (a) To run a read-only audit script on the laboratory machines that records their current security configuration only. This script reads configuration values and does not access, collect, or modify any user data.
- (b) To deploy a set of standard, documented security policies to the pilot laboratory through Group Policy or equivalent management, including disabling browser sign-in and synchronisation, disabling the browser password manager, enforcing ephemeral browser profiles, and restricting extension installation.
- (c) To enable Windows Shared PC Mode (or the Unified Write Filter) on the pilot laboratory so that user profile data does not persist between sessions.

- (d) To enable Microsoft Defender Attack Surface Reduction rules, first in audit mode and then in block mode after a short review period, together with standard application-control measures.
- (e) To install the free monitoring tools Microsoft Sysmon and Wazuh on the pilot laboratory for the purpose of detecting attempted attacks, configured to record process and network metadata only, and not the contents of any user's work, documents, or communications.
- (f) Subject to the approval of the appropriate University ethics review, to conduct an anonymous survey of laboratory users and to perform controlled credential-extraction tests using only test accounts created and owned by the research team.

3. Safeguards and Commitments

The following commitments will be observed at every stage of the work, without exception:

- At no point will the research team read, store, copy, or transmit any real credential or personal data belonging to any student or member of staff. All credential-extraction testing will use only test accounts created by the research team for this purpose.
- The anonymous survey will collect no names, no student identification numbers, and no information by which any individual could be identified.
- All work will be performed under the supervision of, and with the cooperation of, the IT Office. No configuration of any University machine will be altered unilaterally or without prior agreement.
- Notices will be displayed in the affected laboratory informing users that a security configuration pilot is in progress.
- The monitoring tools will be limited to security-relevant process and network metadata, and a clear statement of what is and is not monitored will be made available.

4. Limited Risk and Full Reversibility

Every configuration change described above is applied through standard management mechanisms and is fully reversible within minutes: removing the relevant policy returns the machine to its previous state at the next user login. The monitoring tools observe only and do not block normal use. The control laboratory remains entirely unchanged throughout. Should any change cause unexpected disruption to legitimate coursework or University operations, it can be reverted immediately and at the discretion of the IT Office. The pilot involves no purchase of software or hardware and imposes no licensing cost on the University.

5. Outcome and Deliverables

On completion, I will provide the IT Office with a written report describing the measured security improvement, the configuration applied, and recommendations for any wider adoption the University may wish to consider. With the University's consent, anonymised and aggregated findings may also form the basis of an academic research publication, conducted under the supervision of Dr. Nova Ahmed and subject to ethics approval, which would contribute a documented, reproducible solution of value to other institutions facing the same problem.

I would be grateful for the opportunity to discuss this request at your convenience and to provide any further detail the IT Office may require. I am committed to conducting this work responsibly, transparently, and entirely within the scope of the authorization granted.

Respectfully submitted,

Khondokar Sajid

ID 2211954042 | Department of CSE, North South University

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Endorsement of Supervisor

I confirm that I am supervising this research and endorse the request above.

Dr. Nova Ahmed | Department of CSE | DIAL Lab, North South University

For Office Use — Authorization

Approved / Not approved (please circle). Conditions, if any:

Director, Information Technology Office | Date: _____